

Solutions to Norris' **Markov Chains**

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I was working through *Markov Chains*, Cambridge University Press, 1997 by J.R. Norris and decided that I might as well provide my solutions in the off-chance that they are of use to others. If interested, see olivermaynard.com for more about me and the things that I do :)

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1 Discrete-time Markov chains

1.1 Definition and Basic Properties

Exercise 1.1.1. Let $B_1, B_2, \dots$ be disjoint events with $\bigcup_{n=1}^{\infty} B_n = \Omega$. Suppose A is another event with $\mathbb{P}(A | B_n) = p$ for every n . Show that $\mathbb{P}(A) = p$. Deduce that if X and Y are discrete random variables, then the following are equivalent:

- (a) X and Y are independent;
- (b) the conditional distribution of X given $Y = y$ is independent of y .

Proof. Consider first that

$$\mathbb{P}(A | B_n) = p \quad \text{for } n = 1, 2, \dots,$$

and that by the law of total probability,

$$\mathbb{P}(A) = \sum_{n=1}^{\infty} \mathbb{P}(A | B_n) \mathbb{P}(B_n) = \sum_{n=1}^{\infty} p \mathbb{P}(B_n).$$

Since the B_n are disjoint with $\bigcup_{n=1}^{\infty} B_n = \Omega$, we have $\sum_{n=1}^{\infty} \mathbb{P}(B_n) = \mathbb{P}(\Omega) = 1$, and

$$\sum_{n=1}^{\infty} p \mathbb{P}(B_n) = p \sum_{n=1}^{\infty} \mathbb{P}(B_n) = p(1) = p.$$

So

$$\mathbb{P}(A) = p.$$

□

We now demonstrate the equivalence of (a) and (b).

(a) $\Rightarrow$ (b). If X and Y are independent, then

$$\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x) \mathbb{P}(Y = y) \quad \text{for all } x, y.$$

Now consider that, as a direct result,

$$\mathbb{P}(X = x | Y = y) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)} = \mathbb{P}(X = x).$$

Since this is true for every value x and every y , the conditional distribution of X given $Y = y$ is, for each y , exactly the distribution of X ; i.e.,

$$\mathbb{P}(X | Y = y) = \mathbb{P}(X) \quad \text{for all } y,$$

which is statement (b).

(b) $\Rightarrow$ (a). If the conditional distribution of X given $Y = y$ is independent of y , then

$$\mathbb{P}(X | Y = y) = \mu(X) \quad \text{for all } y \text{ (with } \mathbb{P}(Y = y) > 0),$$

where $\mu(X)$ is some generic distribution. Since $\mu(X)$ is constant for all y , we may use the above result regarding $\mathbb{P}(A | B_n) = p$ and obtain

$$\mathbb{P}(X) = \mu(X) \implies \mathbb{P}(X | Y = y) = \mathbb{P}(X).$$

So

$$\underbrace{\mathbb{P}(X = x | Y = y)}_{= \mathbb{P}(X=x)} = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)} \implies \mathbb{P}(X = x) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)}.$$

Therefore,

$$\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x) \mathbb{P}(Y = y) \quad \text{for all } x, y,$$

which is exactly the definition of independence between X and Y .

Exercise 1.1.2. Suppose that $(X_n)_{n \geq 0}$ is Markov(λ, P). If $Y_n = X_{kn}$, show that $(Y_n)_{n \geq 0}$ is Markov(λ, P^k).

Proof. Since $Y_n = X_{kn}$, we observe that

$$\mathbb{P}(Y_0 = i_0, \dots, Y_n = i_n) = \mathbb{P}(X_0 = i_0, X_k = i_1, \dots, X_{kn} = i_n).$$

By the Markov property, this probability factors as

$$\mathbb{P}(X_0 = i_0, X_k = i_1, \dots, X_{kn} = i_n) = \lambda_{i_0} p_{i_0 i_1}^* p_{i_1 i_2}^* \cdots p_{i_{n-1} i_n}^*,$$

where the p_{ij}^* are the transition probabilities of the chain $(X_n)_{n \geq 0}$ from a state i at X_m (for any integer $m \geq 0$) to a state j at X_{m+k} , i.e., k steps of the chain later. Formally, this probability is

$$p_{ij}^* = \mathbb{P}_i(X_{m+k} = j) = \mathbb{P}(X_{m+k} = j \mid X_m = i),$$

which was shown in Theorem 1.1.3(ii) to be exactly $p_{ij}^{(k)}$, the (i, j) entry in P^k . So

$$\mathbb{P}(Y_0 = i_0, \dots, Y_n = i_n) = \lambda_{i_0} p_{i_0 i_1}^{(k)} p_{i_1 i_2}^{(k)} \cdots p_{i_{n-1} i_n}^{(k)},$$

and by Theorem 1.1.1, this is exactly the statement that $(Y_n)_{n \geq 0}$ is Markov(λ, P^k). This result makes intuitive sense, since we may interpret each step and transition probability between states in the chain $(Y_n)_{n \geq 0}$ as corresponding to a k -step journey between the same states in $(X_n)_{n \geq 0}$. $\square$

Exercise 1.1.3. Let X_0 be a random variable with values in a countable set I . Let $Y_1, Y_2, \dots$ be a sequence of independent random variables, uniformly distributed on $[0, 1]$. Suppose we are given a function

$$G : I \times [0, 1] \rightarrow I$$

and define inductively

$$X_{n+1} = G(X_n, Y_{n+1}).$$

Show that $(X_n)_{n \geq 0}$ is a Markov chain and express its transition matrix P in terms of G . Can all Markov chains be realized in this way? How would you simulate a Markov chain using a computer?

Suppose now that $Z_0, Z_1, \dots$ are independent, identically distributed random variables such that $Z_i = 1$ with probability p and $Z_i = 0$ with probability $1 - p$. Set $S_0 = 0$, $S_n = Z_1 + \cdots + Z_n$. In each of the following cases, determine whether $(X_n)_{n \geq 0}$ is a Markov chain:

- (a) $X_n = Z_n$;
- (b) $X_n = S_n$;
- (c) $X_n = S_0 + \cdots + S_n$;
- (d) $X_n = (S_n, S_0 + \cdots + S_n)$.

In the cases where $(X_n)_{n \geq 0}$ is a Markov chain, find its state-space and transition matrix; in the cases where it is not, give an example where $\mathbb{P}(X_{n+1} = i \mid X_n = j, X_{n-1} = k)$ is not independent of k .

Proof. We first consider the probability of the intersection

$$\mathbb{P}(X_0 = i_0, \dots, X_n = i_n) = \mathbb{P}(X_0 = i_0) \cdot \mathbb{P}(X_1 = i_1 \mid X_0 = i_0) \cdots \mathbb{P}(X_n = i_n \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1})$$

where the conditional probabilities are each given by (using an arbitrary index m in the chain)

$$\mathbb{P}(X_{m+1} = i_{m+1} \mid X_0 = i_0, \dots, X_m = i_m) = \mathbb{P}(G(X_m, Y_{m+1}) = i_{m+1} \mid X_0 = i_0, \dots, X_m = i_m).$$

Since $X_0, \dots, X_m$ are determined entirely by X_0 and $Y_1, \dots, Y_m$, and Y_{m+1} is independent of $Y_1, \dots, Y_m$ (and of X_0), it follows that Y_{m+1} is independent of $X_0, \dots, X_m$. Thus

$$\mathbb{P}(G(X_m, Y_{m+1}) = i_{m+1} \mid X_0 = i_0, \dots, X_m = i_m) = \mathbb{P}(G(i_m, Y_{m+1}) = i_{m+1}).$$

Further, observe that since $Y_1, Y_2, Y_3, \dots$ are independent and identically distributed uniformly in $[0, 1]$, $G(i, \cdot)$ is computed the same regardless of which Y_{m+1} you plug in (i.e., which time index you happen to be

looking at). For any transition from state i to j this means we have

$$\mathbb{P}(G(i, Y_{m+1}) = j) = \mathbb{P}(G(i, Y) = j) \text{ where } Y \sim \text{Uniform}[0, 1],$$

and can then define the transition probabilities in the natural way

$$p_{ij} := \mathbb{P}(G(i, Y) = j).$$

We can finally write

$$\mathbb{P}(X_0 = i_0, \dots, X_n = i_n) = \lambda_{i_0} p_{i_0 i_1} p_{i_1 i_2} \cdots p_{i_{n-1} i_n},$$

and by Theorem 1.1.1 conclude that $(X_n)_{n \geq 0}$ with $X_0 \sim \lambda$ is therefore Markov(λ, P), with the (i, j) entry of the transition matrix given by the corresponding transition probability p_{ij} . $\square$

Remarkably, *any* Markov chain can be realized in this way. To see this, fix a state i . The row $(p_{ij})_{j \in I}$ of P is a probability distribution over I . Since I is countable, we can list its elements $j_1, j_2, j_3, \dots$ and partition $[0, 1]$ into consecutive intervals

$$A_{ij_1} = [0, p_{ij_1}), \quad A_{ij_2} = [p_{ij_1}, p_{ij_1} + p_{ij_2}), \quad A_{ij_3} = [p_{ij_1} + p_{ij_2}, p_{ij_1} + p_{ij_2} + p_{ij_3}), \dots$$

each having exactly length p_{ij_k} . We then define

$$G(i, u) = j_k \quad \text{whenever } u \in A_{ij_k}.$$

Observe that for U uniform on $[0, 1]$,

$$\mathbb{P}(G(i, U) = j_k) = \mathbb{P}(U \in A_{ij_k}) = \text{length of } A_{ij_k} = p_{ij_k}$$

which exactly matches the prescribed transition probability. So the chain $X_{n+1} = G(X_n, Y_{n+1})$ built from this G has exactly transition matrix P (*any* P you started with). This construction is exactly the way you would simulate a Markov chain on a computer. Given you're in state i , draw a uniform random number U and use the cumulative sums of the row p_i to decide which "bucket" U falls into (which tells you the next state). The G function is in essence just the inverse cumulative density function of row i of P .

Classifying Random Processes:

- (a) We consider $X_n = Z_n$ and

$$\begin{aligned} \mathbb{P}(X_0 = i_0, \dots, X_n = i_n) &= \mathbb{P}(X_0 = i_0) \cdot \mathbb{P}(X_1 = i_1 \mid X_0 = i_0) \cdots \\ &\quad \mathbb{P}(X_n = i_n \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}). \end{aligned}$$

Since $X_n = Z_n$, and $Z_0, Z_1, \dots$ are i.i.d., X_n is independent of $X_0, \dots, X_{n-1}$ entirely. In particular,

$$\mathbb{P}(X_n = i_n \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}) = \mathbb{P}(X_n = i_n) = \mathbb{P}(Z_n = i_n),$$

which satisfies the (trivial) Markov property

$$\mathbb{P}(X_n = i_n \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}) = \mathbb{P}(X_n = i_n \mid X_{n-1} = i_{n-1}),$$

since neither side depends on any of $i_0, \dots, i_{n-2}$, or even on i_{n-1} . Substituting this into the original factorization gives

$$\mathbb{P}(X_0 = i_0, \dots, X_n = i_n) = \lambda_{i_0} p_{i_0 i_1} \cdots p_{i_{n-1} i_n},$$

where $p_{ij} := \mathbb{P}(Z = j)$ for any i , and thus $(X_n)_{n \geq 0}$ is Markov(λ, P). Z only takes on values 0 and 1, so the state-space I is just the set $\{0, 1\}$, with the transition matrix and initial distribution given directly by the problem:

$$P = \begin{pmatrix} p & 1-p \\ p & 1-p \end{pmatrix}, \quad \lambda = \begin{pmatrix} p \\ 1-p \end{pmatrix}.$$

Observe that the rows of P are identical, since the transition probability from *any* state i to state j does not depend on i .

- (b) We now take under consideration $X_n = S_n = Z_1 + \cdots + Z_n$ with $X_0 = S_0 = 0$, and observe for a general index in the chain

$$\mathbb{P}(X_n = i_n \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}).$$

Conditioning on $X_0 = 0, X_1 = i_1, \dots, X_{n-1} = i_{n-1}$ is equivalent to conditioning on the individual values $Z_1 = i_1, Z_2 = i_2 - i_1, \dots, Z_{n-1} = i_{n-1} - i_{n-2}$, since each X_m determines Z_m once X_{m-1} is also known. Writing $X_n = X_{n-1} + Z_n = i_{n-1} + Z_n$, we have

$$\mathbb{P}(X_n = i_n \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}) = \mathbb{P}(i_{n-1} + Z_n = i_n \mid Z_1 = i_1, \dots, Z_{n-1} = i_{n-1} - i_{n-2}).$$

Since $Z_1, Z_2, \dots$ are i.i.d., Z_n is independent of $Z_1, \dots, Z_{n-1}$, so conditioning on the earlier Z 's gives no information about Z_n , and the conditional collapses to

$$\mathbb{P}(X_n = i_n \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}) = \mathbb{P}(i_{n-1} + Z_n = i_n) = \mathbb{P}(Z_n = i_n - i_{n-1}) =: p_{i_{n-1}i_n}.$$

This depends only on i_{n-1} and i_n , not on $i_0, \dots, i_{n-2}$, so

$$\mathbb{P}(X_n = i_n \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}) = \mathbb{P}(X_n = i_n \mid X_{n-1} = i_{n-1}) = p_{i_{n-1}i_n},$$

which is exactly the Markov property. Thus,

$$\mathbb{P}(X_0 = i_0, \dots, X_n = i_n) = \lambda_{i_0} p_{i_0i_1} \cdots p_{i_{n-1}i_n},$$

and $(X_n)_{n \geq 0}$ is a Markov chain. Since $X_n = Z_1 + \cdots + Z_n$ and each $Z_i \in \{0, 1\}$, the sum can take any non-negative integer value, increasing by at most 1 each step starting from $X_0 = 0$. So $I = \{0, 1, 2, \dots\} = \mathbb{Z}_{\geq 0}$, the non-negative integers. The transition matrix follows from the definition of each $p_{i_{n-1}i_n}$:

$$p_{ij} = \begin{cases} 1 - p & \text{if } j = i, \\ p & \text{if } j = i + 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (c) The chain $X_n = S_0 + \cdots + S_n$ with $X_0 = S_0 = 0$ is not a Markov chain. We construct an example where $\mathbb{P}(X_{n+1} = i \mid X_n = j, X_{n-1} = k)$ is not independent of k . Consider two paths leading to the same value ($j = 3$) of X_3 (i.e. $n = 3$):

- Path $(Z_1, Z_2, Z_3) = (1, 0, 0)$ gives $X_2 = S_1 + S_2 = 2$ and $X_3 = X_2 + S_3 = 3$, so $k = X_2 = 2$.
- Path $(Z_1, Z_2, Z_3) = (0, 1, 1)$ gives $X_2 = S_1 + S_2 = 1$ and $X_3 = X_2 + S_3 = 3$, so $k = X_2 = 1$.

Considering $X_4 = 5$ (a value reachable from both branches, via opposite outcomes of Z_4) in the former case

$$\mathbb{P}(X_4 = 5 \mid X_3 = 3, X_2 = 2) = \mathbb{P}(Z_4 = 1) = p,$$

since this path forces $S_3 = 1$, giving $X_4 = 4 + Z_4$, so $X_4 = 5 \Leftrightarrow Z_4 = 1$. In the latter case,

$$\mathbb{P}(X_4 = 5 \mid X_3 = 3, X_2 = 1) = \mathbb{P}(Z_4 = 0) = 1 - p,$$

since this path forces $S_3 = 2$, giving $X_4 = 5 + Z_4$, so $X_4 = 5 \Leftrightarrow Z_4 = 0$. These two conditional probabilities differ whenever $p \neq \frac{1}{2}$, so $\mathbb{P}(X_{n+1} = i \mid X_n = j, X_{n-1} = k)$ is not independent of k , contradicting the Markov property. Hence $(X_n)_{n \geq 0}$ is not a Markov chain.

- (d) Recall that this issue with the chain in problem (c) was that $X_n = S_0 + \cdots + S_n$ alone doesn't tell you S_n specifically, and so different paths can give the same cumulative sum but different "current" S_n (which informs the next increment).

Here, we have a pair $X_n = (S_n, S_0 + \cdots + S_n)$. We can write $X_n = (S_n, T_n)$ where $T_n := S_0 + \cdots + S_n$. Then

$$S_{n+1} = S_n + Z_{n+1}, \quad T_{n+1} = T_n + S_{n+1} = T_n + S_n + Z_{n+1}.$$

So $X_{n+1} = (S_{n+1}, T_{n+1}) = (S_n + Z_{n+1}, T_n + S_n + Z_{n+1})$, a function of X_n and the fresh randomness Z_{n+1} alone (exactly the structure of the general $X_{n+1} = G(X_n, Y_{n+1})$ with i.i.d. driving the randomness we saw before!), so the chain satisfies the Markov property, and therefore *is* Markov. The state-space is

$$I = \{(s, t) \in \mathbb{Z}_{\geq 0} \times \mathbb{Z}_{\geq 0} : t \geq s\}.$$

The probabilities of transition $G((s, t), z) = (s + z, t + s + z)$ with $Z \in \{0, 1\}$ are

$$p_{(s,t),(s',t')} = \begin{cases} 1 - p & \text{if } (s', t') = (s, t + s), \\ p & \text{if } (s', t') = (s + 1, t + s + 1), \\ 0 & \text{otherwise.} \end{cases}$$